General Linear Model

Lecture Notes

A rigorous treatment of the General Linear Model $(y, X\beta, \sigma^2 V)$, covering estimability, BLUE, hypothesis testing, variance component estimation, and the MINQUE procedure.

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Chapter 1

Introduction to the General Linear Model

1.1 From the Standard Model to the General Model

In the *standard* linear model we write

$$(y,\ X\beta,\ \sigma^2 I),$$

meaning $y = X\beta + \varepsilon$ with $\mathrm{E}(\varepsilon) = 0$ and $\mathcal{D}(\varepsilon) = \sigma^2 I$. The **General Linear Model** replaces the identity covariance structure with an arbitrary (possibly singular) known matrix V:

$$\boxed{(y,\ X\beta,\ \sigma^2 V), \qquad V \text{ known, possibly singular.}}$$

1.2 Motivation: Why Do We Need It?

1.2.1 Linear Constraints on Parameters

Consider the standard model $y = X\beta + \varepsilon$, $\mathcal{D}(\varepsilon) = \sigma^2 I$, subject to the linear constraint $A\beta = \xi \in \mathcal{C}(A)$. Augmenting the observation vector:

$$y_* = \begin{pmatrix} y \\ \xi \end{pmatrix} = \begin{pmatrix} X \\ A \end{pmatrix} \beta + \varepsilon_*, \qquad \mathcal{D}(\varepsilon_*) = \sigma^2 \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix},$$

which is a General Linear Model with $V = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}$ (singular).

1.2.2 Reduction of Nuisance Parameters

Write $y = X_1\beta_1 + X_2\beta_2 + \varepsilon$ where $X_2\beta_2$ is a nuisance component. Projecting out $\mathcal{C}(X_2)$:

$$(I - P_{X_2})y = (I - P_{X_2})X_1\beta_1 + \varepsilon,$$

and the residual covariance becomes $\sigma^2(I - P_{X_2})$, again a General Linear Model.

1.3 Model Consistency

Example 1.1. Consider the 4×1 system

$$y = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad \sigma^2 = 1, \quad V = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here V projects out rows 2 and 4, meaning those components are *deterministic*. Checking consistency requires verifying that the data satisfy $y - X\beta \in \mathcal{C}(V)$ almost surely (a.s.).

The model $y = X\beta + \varepsilon$ with $\mathcal{D}(\varepsilon) = \sigma^2 V$ and $\mathrm{E}(\varepsilon) = 0$ implies

$$y - X\beta \in \mathcal{C}(V) \text{ a.s.}, \qquad \text{hence} \qquad y \in \mathcal{C}(X : V) \text{ a.s.}$$

Theorem 1.1 (Column-space decomposition)**.** *$\mathcal{C}(X : V)$ admits two orthogonal decompositions:*

$$\mathcal{C}(X : V) = \mathcal{C}(X) \oplus \mathcal{C}\big((I - P_X)V\big) \tag{a}$$
$$\mathcal{C}(X : V) = \mathcal{C}(V) \oplus \mathcal{C}\big((I - P_V)X\big) \tag{b}$$

Proof. For decomposition (a), take any $u \in \mathcal{C}(X : V)$; write $u = Xc + Vd$ for some vectors c, d. Then $(I - P_X)u = (I - P_X)Vd \in \mathcal{C}((I - P_X)V)$, and $P_X u \in \mathcal{C}(X)$. These two components are orthogonal since $\mathcal{C}(X) \perp \mathcal{C}((I - P_X)V)$ by construction of $I - P_X$. Conversely, if $u = Xa + (I - P_X)Vb$, then $u \in \mathcal{C}(X : V)$. Decomposition (b) follows symmetrically. ∎

As a consequence, $y \in \mathcal{C}(X : V)$ if and only if

$$(I - P_X)y \in \mathcal{C}\big((I - P_X)V\big), \tag{b}$$

$$(I - P_V)y \in \mathcal{C}\big((I - P_V)X\big). \tag{a}$$

Note 1.2. $\mathcal{D}\big((I - P_V)y\big) = \sigma^2(I - P_V)V = 0$, so $(I - P_V)y$ is *non-stochastic*. We write $d := (I - P_V)y$ (a fixed vector), and any admissible β must satisfy

$$(I - P_V)X\beta = d.$$

Chapter 2

Linear Unbiased Estimation

2.1 Linear Unbiased Estimators (LUEs)

Definition 2.1 (LUE in the General Linear Model)**.** In the model $(y, X\beta, \sigma^2 V)$, a linear statistic $k^T y$ is a **Linear Unbiased Estimator (LUE)** of the linear parametric function $p^T\beta$ if

$$\mathrm{E}(k^T y) = p^T\beta \qquad \forall\, \beta \in \mathbb{R}^{p+1} \text{ satisfying } (I - P_V)X\beta = d.$$

Note 2.2. In the standard model $(y, X\beta, \sigma^2 I)$, the condition reduces to $X^T\ell = p$: the vector ℓ must lie in a row of X. In the general model, this is modified.

2.1.1 Constructing LUEs

Suppose ℓ is such that $X^T\ell = p$. Then $\ell^T y$ is an LUE of $p^T\beta$. For any vector m, define

$$k := \ell + (I - P_V)(I - P_d)m,$$

where P_d is the orthogonal projector onto $\mathcal{C}(d)$. Then

$$k^T y = \ell^T y + m^T(I - P_d)(I - P_V)y.$$

Since $(I - P_V)y = d$ is fixed, the added term $(I - P_d)(I - P_V)y = (I - P_d)d = 0$, so $k^T y = \ell^T y$ a.s. and $k^T y$ is also an LUE of $p^T\beta$.

Example 2.1. With the earlier data

$$V = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad d = (I - P_V)y = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 4 \end{bmatrix},$$

$$(I - P_V)(I - P_d) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0.8 & 0 & -0.4 \\ 0 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0.2 \end{bmatrix}.$$

To estimate $p^T\beta = \beta_0 + \beta_1$, take $\ell = (1, 0, 0, 0)^T$ (so $X^T\ell = (1, 1, 0)^T = p$) and $m = (0, 0, 0, 1)^T$. Then

$$(I - P_V)(I - P_d)m = \begin{bmatrix} 0 \\ -0.4 \\ 0 \\ 0.2 \end{bmatrix}, \quad k = \begin{bmatrix} 1 \\ -0.4 \\ 0 \\ 0.2 \end{bmatrix}.$$

We compute

$$X^T k = \begin{bmatrix} 1 - 0.4 + 0.2 \\ 1 - 0.4 + 0 \\ 0 + 0.2 \end{bmatrix} = \begin{bmatrix} 0.8 \\ 0.6 \\ 0.2 \end{bmatrix} \neq p,$$

illustrating that $X^T k \neq p$ in general. However, $k^T y = \ell^T y = 1$ a.s.

2.2 Characterisation of LUEs

Theorem 2.3 (LUE Characterisation)**.** *In $(y, X\beta, \sigma^2 V)$, $k^T y$ is an LUE of $p^T\beta$ if and only if there exists ℓ such that*

$$k^T y = \ell^T y \text{ a.s.} \quad \text{and} \quad X^T \ell = p.$$

Proof. **(⇐)** If $X^T\ell = p$, then $\mathrm{E}(\ell^T y) = \ell^T X\beta = p^T\beta$ for all admissible β, so $\ell^T y$ is an LUE. Since $k^T y = \ell^T y$ a.s., $k^T y$ is also an LUE.

(⇒) Suppose $k^T y$ is an LUE of $p^T\beta$. Write

$$k = \ell + (I - P_V)(I - P_d)m, \quad \ell = k - (I - P_V)(I - P_d)m.$$

Then $X^T\ell = X^T k - X^T(I - P_V)(I - P_d)m$, so we need

$$X^T k - p \in \mathcal{C}\big(X^T(I - P_V)(I - P_d)\big).$$

To establish this, let $w \neq 0$ satisfy $w^T\big(X^T(I - P_V)(I - P_d)\big)^T w = 0$, i.e. $(I - P_d)(I - P_V)Xw = 0$. We show $w^T(X^T k - p) = 0$ by considering two cases.

Case 1: There exists $\alpha_1 \neq 0$ with $(I - P_V)Xw = \alpha_1 d$. Setting $\alpha = 1/\alpha_1$, we have $(I - P_V)X(\alpha w) = d$, so αw is an admissible β. Since $k^T y$ is an LUE:

$$k^T X(\alpha w) = p^T(\alpha w) \implies (X^T k - p)^T w = 0.$$

Case 2: $(I - P_V)Xw = 0$. If β_0 solves $(I - P_V)X\beta = d$, then $\beta_1 = \beta_0 + w$ also solves it. From the LUE condition applied to both:

$$k^T X\beta_0 = p^T\beta_0, \quad k^T X\beta_1 = p^T\beta_1,$$

subtracting gives $k^T Xw = p^T w$, hence $(X^T k - p)^T w = 0$. ■

2.3 Linear Zero Functions (LZFs)

Definition 2.4 (LZF)**.** A linear statistic $k^T y$ is a **Linear Zero Function (LZF)** if $\mathrm{E}(k^T y) = 0$ for all admissible β.

Corollary 2.5 (LZF Characterisation)**.** *In $(y, X\beta, \sigma^2 V)$, $k^T y$ is an LZF if and only if there exists ℓ such that*

$$k^T y = \ell^T y \text{ a.s.} \quad \text{and} \quad X^T\ell = 0.$$

Corollary 2.6 (LZF as function of residual)**.** *Any LZF can be written as a linear function of $(I - P_X)y$ a.s.*

Proof. $k^T y$ is an LZF $\Rightarrow \exists\, \ell$ with $X^T \ell = 0 \Rightarrow \ell \in \mathcal{C}(I - P_X) \Rightarrow \exists\, m$ with $\ell = (I - P_X)m$ $\Rightarrow k^T y = \ell^T y = m^T (I - P_X) y$ a.s. ■

2.4 Estimability

The same characterisation of estimability as in the standard model continues to hold.

Theorem 2.7 (Estimability)**.** *In* $(y, X\beta, \sigma^2 V)$*:*

(i) $p^T\beta$ *is estimable* $\iff$ $p \in \mathcal{C}(X^T)$*.*

(ii) $p^T\beta$ *is estimable* $\iff$ $p^T\beta$ *is identifiable.*

Proof. **(i) (⇐):** If $p \in \mathcal{C}(X^T)$, then $p = X^T \ell$ for some ℓ, and $\ell^T y$ is an LUE.

(i) (⇒): If $p^T\beta$ is estimable, by Theorem 2.3 there exists ℓ with $X^T \ell = p$, so $p \in \mathcal{C}(X^T)$.

(ii) Left as Homework Exercise 2.4. ■

Homework Exercise. Prove part (ii) of Theorem 2.7: $p^T\beta$ is estimable if and only if it is identifiable.

Solution. Recall that $p^T\beta$ is *identifiable* if $p^T\beta_1 = p^T\beta_2$ for all pairs (β_1, β_2) that are observationally equivalent, i.e., $X\beta_1 = X\beta_2$.

Estimable ⇒ Identifiable: If $X\beta_1 = X\beta_2$ then $X(\beta_1 - \beta_2) = 0$, so $\beta_1 - \beta_2 \in \ker(X)$. Since $p \in \mathcal{C}(X^T)$, we have $p \perp \ker(X)$, so $p^T(\beta_1 - \beta_2) = 0$, giving $p^T\beta_1 = p^T\beta_2$.

Identifiable ⇒ Estimable: Suppose $p \notin \mathcal{C}(X^T)$. Then there exists $w \in \ker(X^T)^\perp = \ker(X)$ with $p^T w \neq 0$. For any admissible β_0, the vector $\beta_1 = \beta_0 + w$ is also admissible with $X\beta_1 = X\beta_0$, yet $p^T\beta_1 \neq p^T\beta_0$. Contradiction. ■

Chapter 3

Best Linear Unbiased Estimation (BLUE)

3.1 Definition

Definition 3.1 (BLUE)**.** In $(y, X\beta, \sigma^2 V)$, $k^T y$ is the **Best Linear Unbiased Estimator (BLUE)** of an estimable $p^T\beta$ if:

(a) $k^T y$ is an LUE of $p^T\beta$, and

(b) $\mathrm{Var}(k_1^T y) \geq \mathrm{Var}(k^T y)$ for every other LUE $k_1^T y$ of $p^T\beta$.

3.2 Characterisation via LZFs

Theorem 3.2 (BLUE–LZF Characterisation)**.** *$k^T y$ is the BLUE of $p^T\beta$ if and only if it is uncorrelated with every LZF.*

Proof. **(⇒)** Suppose for contradiction that there exists a non-trivial LZF $a^T y$ with $\mathrm{Cov}(k^T y,\, a^T y) \neq 0$. Define

$$\ell_1^T y \;:=\; k^T y - \frac{\mathrm{Cov}(k^T y,\, a^T y)}{\mathrm{Var}(a^T y)}\, a^T y.$$

This is still an LUE of $p^T\beta$ (subtracting an LZF preserves unbiasedness). Its variance is

$$\mathrm{Var}(\ell_1^T y) = \mathrm{Var}(k^T y) - \frac{[\mathrm{Cov}(k^T y, a^T y)]^2}{\mathrm{Var}(a^T y)} < \mathrm{Var}(k^T y),$$

contradicting the BLUE property of $k^T y$.

($\Leftarrow$) Let $k_1^T y$ be any LUE of $p^T\beta$. Write

$$k_1^T y = k^T y + \underbrace{(k_1 - k)^T y}_{\text{LZF}}.$$

Since $k^T y$ is uncorrelated with the LZF $(k_1 - k)^T y$:

$$\mathrm{Var}(k_1^T y) = \mathrm{Var}(k^T y) + \mathrm{Var}((k_1 - k)^T y) \geq \mathrm{Var}(k^T y).$$

■

Remark 3.3. The proof of ($\Rightarrow$) also shows that the BLUE of $p^T\beta$ is *unique*: if $k^T y$ and $k_1^T y$ are both BLUEs, their difference is an LZF uncorrelated with itself, hence zero.

3.3 Explicit Formula for the BLUE

Theorem 3.4 (BLUE Formula)**.** *Suppose $A\beta$ is estimable, so $\mathcal{C}(A^T) \subseteq \mathcal{C}(X^T)$ and $A = LX$ for some L. The unique BLUE of $A\beta = LX\beta$ is*

$$\widehat{LX\beta} \;=\; L\Big[I - V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)\Big]y.$$

In particular, taking $L = I$:

$$\boxed{\hat{y} = \Big[I - V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)\Big]y,}$$

and the residual vector is

$$e = y - \hat{y} = V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)y.$$

Taking $L = AX^{-}$ gives $\widehat{A\beta} = A\hat{\beta}$ where $\hat{\beta} = X^{-}\hat{y}$.

Proof. We seek a modification of any LUE Ly that is uncorrelated with $(I - P_X)y$, the generator of all LZFs. By the following lemma this is achieved by subtracting $B(I - P_X)y$ with B chosen via covariance projection.

Key Lemma. *Let U and W be random vectors with finite second moments and* $\mathrm{E}(W) = 0$.

Then $U - BW$ is uncorrelated with W if and only if

$$BW = \mathrm{Cov}(U, W)\,[\mathcal{D}(W)]^{-}W.$$

Proof of Lemma. ($\Leftarrow$): Direct computation gives

$$\mathrm{Cov}(U - BW, W) = \mathrm{Cov}(U, W) - B\mathcal{D}(W) = \mathrm{Cov}(U, W) - \mathrm{Cov}(U, W)\mathcal{D}(W)^{-}\mathcal{D}(W) = 0,$$

using the property of Moore–Penrose pseudoinverse.

($\Rightarrow$): $\mathrm{Cov}(U - BW, W) = 0 \Rightarrow \mathrm{Cov}(U, W) = B\mathcal{D}(W) \Rightarrow B^T = [\mathcal{D}(W)]^{-}\,\mathrm{Cov}(W, U) \Rightarrow BW = \mathrm{Cov}(U, W)[\mathcal{D}(W)]^{-}W$. ■

Applying the lemma with $U = Ly$ and $W = (I - P_X)y$:

$$\mathrm{Cov}(Ly, (I - P_X)y) = L\sigma^2 V(I - P_X),$$

$$\mathcal{D}((I - P_X)y) = \sigma^2(I - P_X)V(I - P_X).$$

Hence the optimal B satisfies

$$B(I - P_X)y = LV(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)y,$$

and the BLUE of $LX\beta$ is

$$Ly - B(I - P_X)y = L\Big[I - V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)\Big]y. \qquad ■$$

3.4 Dispersion of the BLUE and Residuals

Theorem 3.5 (Dispersion Formulas)**.** *In the General Linear Model $(y, X\beta, \sigma^2 V)$:*

$$\mathcal{D}(\hat{y}) = \sigma^2\Big[I - V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)\Big]V, \tag{3.1}$$

$$\mathcal{D}(e) = \sigma^2 V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)V, \tag{3.2}$$

$$\mathcal{D}(y) = \mathcal{D}(\hat{y}) + \mathcal{D}(e). \tag{3.3}$$

Homework Exercise. Prove Theorem 3.5.

Solution. Let $H = V(I - P_X)\big[(I - P_X)V(I - P_X)\big]^{-}(I - P_X)$, so $\hat{y} = (I - H)y$ and $e = Hy$.

Dispersion of $\hat{y}$:

$$\mathcal{D}(\hat{y}) = (I - H)\mathcal{D}(y)(I - H)^T = \sigma^2(I - H)V(I - H)^T.$$

Since $H = V(I - P_X)[(I - P_X)V(I - P_X)]^{-}(I - P_X)$ and $(I - P_X)$ is symmetric idempotent, one can verify $HV = (HV)^T$ and $(I - H)V(I - H)^T = (I - H)V$. Thus

$$\mathcal{D}(\hat{y}) = \sigma^2(I - H)V = \sigma^2\big[I - V(I - P_X)[(I - P_X)V(I - P_X)]^{-}(I - P_X)\big]V.$$

Dispersion of e:

$$\mathcal{D}(e) = H\mathcal{D}(y)H^T = \sigma^2 HVH^T = \sigma^2 HV,$$

using $HV = (HV)^T$ and $H^T = H$. Expanding:

$$\mathcal{D}(e) = \sigma^2 V(I - P_X)[(I - P_X)V(I - P_X)]^{-}(I - P_X)V.$$

Partition:

$$\mathcal{D}(\hat{y}) + \mathcal{D}(e) = \sigma^2(I - H)V + \sigma^2 HV = \sigma^2 V = \mathcal{D}(y).$$

■

Chapter 4

Column Spaces and BLUE-Equivalence

4.1 Column Spaces of the Dispersion Matrices

Theorem 4.1 (Column Space of $\mathcal{D}(\hat{y})$)**.** $\mathcal{C}(\mathcal{D}(\hat{y})) = \mathcal{C}(X) \cap \mathcal{C}(V)$.

Homework Exercise. Prove Theorem 4.1.

Solution. From Theorem 3.5, $\mathcal{D}(\hat{y}) = \sigma^2(I - H)V$ where $H = V(I - P_X)[(I - P_X)V(I - P_X)]^-(I - P_X)$.

($\subseteq$): $(I - H)V = V - HV$ and $HV \in \mathcal{C}(V)$, so $\mathcal{C}(\mathcal{D}(\hat{y})) \subseteq \mathcal{C}(V)$. Also, since $\hat{y} \in \mathcal{C}(X)$ (which follows from $\hat{y} - X\beta \in \mathcal{C}(X) \cap \mathcal{C}(V)$), we have $\hat{y} = P_X\hat{y}$, and applying the dispersion $\mathcal{D}(\hat{y}) = \sigma^2 PX\mathcal{D}(\hat{y})P_X$ gives $\mathcal{C}(\mathcal{D}(\hat{y})) \subseteq \mathcal{C}(P_X) = \mathcal{C}(X)$.

($\supseteq$): Take $u \in \mathcal{C}(X) \cap \mathcal{C}(V)$; write $u = Xa = Vb$. Then $(I - H)u = u - V(I - P_X)[(I - P_X)V(I - P_X)]^-(I - P_X)Xa$. Since $(I - P_X)X = 0$, $(I - H)u = u \in \mathcal{C}(X) \cap \mathcal{C}(V) \subseteq \mathcal{C}(\mathcal{D}(\hat{y}))$. ■

Lemma 4.2 (Column Space of $\mathcal{D}(e)$)**.** $\mathcal{C}(\mathcal{D}(e)) = \mathcal{C}(V(I - P_X))$.

Proof. Let $D_e = \mathcal{D}(e)/\sigma^2 = V(I - P_X)[(I - P_X)V(I - P_X)]^-(I - P_X)V$.

($\subseteq$): $D_e = V(I - P_X) \cdot A$ where $A = [(I - P_X)V(I - P_X)]^-(I - P_X)V$, so $\mathcal{C}(D_e) \subseteq \mathcal{C}(V(I - P_X))$.

($\supseteq$): We show $D_e(I - P_X) = V(I - P_X)$. This requires $\mathcal{C}((I - P_X)V) \subseteq \mathcal{C}((I -$

$P_X)V(I - P_X))$. Note that

$$\mathrm{Cov}((I - P_X)y,\, y) = \sigma^2(I - P_X)V,$$

and the column space inclusion for covariance matrices gives the result. Hence $V(I - P_X) = D_e(I - P_X)$, so $\mathcal{C}(V(I - P_X)) \subseteq \mathcal{C}(D_e)$. ■

4.2 BLUE-Equivalence

Definition 4.3 (BLUE-equivalence)**.** Two models $M_1 : (y, X\beta, \sigma^2 V_1)$ and $M_2 : (y, X\beta, \sigma^2 V_2)$ are **BLUE-equivalent** if the BLUE of every estimable $p^T\beta$ is the same under both models.

Theorem 4.4 (BLUE-equivalence criterion)**.** *M_1 and M_2 are BLUE-equivalent if and only if the spaces of LZFs are the same in both models (assuming both models are consistent).*

Example 4.1 (OLS vs. GLS equivalence). Consider $M_1 : (y, X\beta, \sigma^2 I)$ and $M_2 : (y, X\beta, \sigma^2 V)$. These are BLUE-equivalent if and only if $P_X y$ is the BLUE of $X\beta$ under M_2, which holds if and only if

$$\mathrm{Cov}(P_X y,\, (I - P_X)y) = 0 \text{ under } M_2.$$

Expanding:

$$P_X V(I - P_X) = 0 \iff P_X V = P_X V P_X \iff V P_X = P_X V P_X \quad \text{(by symmetry)} \iff P_X V = V P_X \quad \Leftarrow$$

Chapter 5

Estimation of σ^2

5.1 Standard Basis of LZFs

Definition 5.1 (Standard basis)**.** A vector Z of LZFs is a **standard basis** if its components are uncorrelated with unit variance: $\mathcal{D}(Z) = \sigma^2 I_d$.

Theorem 5.2 (Dimension of the LZF space)**.** *If Z is a standard basis of the LZF space, then the number of elements in Z equals*

$$d = \mathrm{rk}(X : V) - \mathrm{rk}(X).$$

Proof. **Step 1:** $d = \mathrm{rk}(\mathcal{D}(e))$**.**

Both Z and e are generating sets for the space of LZFs, so there exist matrices B $(d \times n)$ and C $(n \times d)$ with $Z = Be$ and $e = CZ$. Hence

$$\mathcal{D}(Z) = B\mathcal{D}(e)B^T = \sigma^2 I_d \implies \mathrm{rk}(B\mathcal{D}(e)B^T) = d \implies \mathrm{rk}(\mathcal{D}(e)) \geq d. \quad (\star_1)$$

Also $\mathcal{D}(e) = \sigma^2 CC^T$, so $\mathrm{rk}(\mathcal{D}(e)) = \mathrm{rk}(CC^T) \leq \mathrm{rk}(C) \leq d.$ $\quad (\star_2)$

Combining $(\star_1)$ and $(\star_2)$: $\mathrm{rk}(\mathcal{D}(e)) = d$.

Step 2: $\mathrm{rk}(\mathcal{D}(e)) = \mathrm{rk}(X : V) - \mathrm{rk}(X)$**.**

By Lemma 4.2, $\mathcal{C}(\mathcal{D}(e)) = \mathcal{C}(V(I - P_X))$, so $\mathrm{rk}(\mathcal{D}(e)) = \mathrm{rk}(V(I - P_X))$. From the column-space decomposition (Theorem 1.1):

$$\mathrm{rk}(X : V) = \mathrm{rk}(X) + \mathrm{rk}((I - P_X)V) = \mathrm{rk}(X) + \mathrm{rk}(V(I - P_X)).$$

Therefore $\mathrm{rk}(\mathcal{D}(e)) = \mathrm{rk}(V(I - P_X)) = \mathrm{rk}(X : V) - \mathrm{rk}(X) = d$. ■

5.2 Distribution of the Error Sum of Squares

The error sum of squares is $R_0^2 = Z^T Z = e^T[(I - P_X)V(I - P_X)]^- e$ (using the standard basis representation).

Theorem 5.3 (Distribution of R_0^2)**.** *Suppose $y \sim N_n(X\beta, \sigma^2 V)$. Then*

$$\frac{R_0^2}{\sigma^2} \sim \chi_d^2, \qquad d = \mathrm{rk}(X : V) - \mathrm{rk}(X).$$

Proof. Since Z is a linear function of y and y is normally distributed, Z is also normal. $\mathrm{E}[Z] = 0$ (it is a basis of LZFs) and $\mathcal{D}(Z) = \sigma^2 I_d$, so $Z \sim N_d(0, \sigma^2 I_d)$. Therefore

$$\frac{1}{\sigma^2} Z^T Z = \sum_{i=1}^{d} \left(\frac{Z_i}{\sigma}\right)^2 \sim \chi_d^2.$$

■

5.3 Unbiased Estimation of σ^2

From Theorem 5.3, $\mathrm{E}[R_0^2] = d\sigma^2$, so the unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{R_0^2}{d}.$$

Chapter 6

Hypothesis Testing

6.1 Testing a Linear Hypothesis

Consider the hypothesis $H_0 : A\beta = \xi$ against $H_1 : A\beta \neq \xi$, where $A\beta$ is estimable.

Under the restricted model M_R (i.e., imposing H_0), the error sum of squares satisfies

$$R_H^2 = R_0^2 + Q, \quad \text{where} \quad Q = (A\hat{\beta} - \xi)^T \left[\frac{\mathcal{D}(A\hat{\beta})}{\sigma^2}\right]^- (A\hat{\beta} - \xi).$$

Theorem 6.1 (Testing $H_0 : A\beta = \xi$). *(i) Q and R_0^2 are independently distributed.*

(ii) Under H_0, $\dfrac{Q}{\sigma^2} \sim \chi^2_{q^*}$ where $q^* = \mathrm{rk}(\mathcal{D}(A\hat{\beta}))$.

Proof. **(i)** R_0^2 is a function of LZFs, and $A\hat{\beta}$ is a BLUE. By Theorem 3.2, any BLUE is uncorrelated with every LZF. Under normality, uncorrelatedness implies independence. Therefore $A\hat{\beta}$ and any LZF are independent, and hence Q and R_0^2 are independent.

(ii) $A\hat{\beta} - \xi$ is a linear function of y with $\mathrm{E}[A\hat{\beta} - \xi] = A\beta - \xi = 0$ under H_0 and $\mathcal{D}(A\hat{\beta} - \xi) = \mathcal{D}(A\hat{\beta}) = \sigma^2\Sigma$ (say). Then $A\hat{\beta} - \xi \sim N(0, \sigma^2\Sigma)$, and by the distribution of quadratic forms in normal vectors:

$$\frac{1}{\sigma^2}(A\hat{\beta} - \xi)^T \Sigma^- (A\hat{\beta} - \xi) \sim \chi^2_{\mathrm{rk}(\Sigma)}.$$

■

Corollary 6.2 (F-Test)**.** *Under $H_0 : A\beta = \xi$, the test statistic*

$$F = \frac{Q/q^*}{R_0^2/d} \sim F_{q^*,\,d}.$$

We reject H_0 at level α if $F > F_{\alpha;\,q^*,\,d}$.

6.2 Testing $H_0 : \sigma^2 = \sigma_0^2$

The statistic $R_0^2/\sigma_0^2 \sim \chi_d^2$ under H_0. We reject when this value falls in the critical region (too large or too small depending on the alternative).

6.3 Special Cases

6.3.1 $V = I$ (Standard OLS)

In $(y, X\beta, \sigma^2 I)$:

$$\hat{y} = P_X y, \qquad R_0^2 = y^T(I - P_X)y, \qquad d = n - \mathrm{rk}(X),$$
$$A\hat{\beta} = A(X^TX)^- X^T y, \qquad \mathcal{D}(A\hat{\beta}) = \sigma^2 A(X^TX)^- A^T.$$

6.3.2 $V > 0$ (Non-singular GLS)

In $(y, X\beta, \sigma^2 V)$ with $V > 0$:

$$\begin{aligned}
R_0^2 &= \min_\beta (y - X\beta)^T V^{-1}(y - X\beta) = (y - X\hat{\beta})^T V^{-1}(y - X\hat{\beta}),\\
X\hat{\beta} &= X(X^TV^{-1}X)^- X^TV^{-1}y,\\
d &= n - \mathrm{rk}(X),\\
\mathcal{D}(A\hat{\beta}) &= \sigma^2 A(X^TV^{-1}X)^- A^T,\\
Q &= (A\hat{\beta} - \xi)^T[A(X^TV^{-1}X)^- A^T]^-(A\hat{\beta} - \xi).
\end{aligned}$$

Chapter 7

The SUR Model and BLUE-Equivalence

7.1 Seemingly Unrelated Regressions

Consider k regression equations

$$y_i = X_i\beta_i + \varepsilon_i, \quad i = 1, \ldots, k,$$

where y_i is $n \times 1$, X_i is $n \times p_i$, $\mathrm{E}[\varepsilon_i] = 0$, $\mathcal{D}(\varepsilon_i) = \sigma_{ii} I_n$, and $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = \sigma_{ij} I_n$.

Stacking into one model:

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix}, \quad X = \begin{bmatrix} X_1 & & \\ & \ddots & \\ & & X_k \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix},$$

$$\mathcal{D}(\varepsilon) = \Sigma \otimes I_n, \quad \Sigma = ((\sigma_{ij}))_{k \times k}.$$

7.2 When Does OLS per Equation Equal BLUE?

The per-equation OLS estimator is $\hat{\beta}_i^{OLS} = (X_i^T X_i)^- X_i^T y_i$. This coincides with the BLUE of the joint model if and only if $(y, X\beta, \sigma^2 I_{nk})$ and $(y, X\beta, \Sigma \otimes I_n)$ are BLUE-equivalent:

$$(\Sigma \otimes I_n)\, \mathcal{C}(X) \subseteq \mathcal{C}(X).$$

For $u \in \mathcal{C}(X)$, $u = (X_1 a_1^T, \ldots, X_k a_k^T)^T$, and

$$(\Sigma \otimes I_n)u = \begin{bmatrix} \sum_j \sigma_{1j} X_j a_j \\ \vdots \\ \sum_j \sigma_{kj} X_j a_j \end{bmatrix}.$$

The condition holds if and only if $\sum_j \sigma_{ij} X_j a_j \in \mathcal{C}(X_i)$ for all i and all a_j. This is satisfied when:

(1) $\sigma_{ij} = 0$ for $i \neq j$ (truly unrelated equations), or

(2) $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \cdots = \mathcal{C}(X_k)$ (all equations share the same column space).

7.3 Common Covariance Structures

(1) **Intra-class correlation:** $V = (1-\rho)I + \rho J$. Common in block designs where units within a block are correlated.

(2) **AR(1) serial correlation:** $V_{ij} = \rho^{|i-j|}$. Common in time-series data with autoregressive errors.

Chapter 8

Variance Component Estimation

8.1 The Variance Component Model

In many applications the covariance matrix is a linear combination of known matrices:

$$V = \sum_{i=1}^{k} \sigma_i^2 V_i,$$

where V_i are known and $\theta_i = \sigma_i^2$ are unknown *variance components*. We wish to estimate $p^T\theta = \sum_i f_i\theta_i$.

Classical methods include:

(1) **Henderson's method** (equating mean squares to expectations).

(2) **MINQUE** (Minimum Norm Quadratic Unbiased Estimation), due to C.R. Rao.

8.2 MINQUE: Setup

Suppose $\varepsilon = U\xi$ where $\mathcal{D}(\xi) = \operatorname{diag}(\sigma_1^2 I_{n_1}, \ldots, \sigma_k^2 I_{n_k})$, so

$$V = \mathcal{D}(\varepsilon) = \sum_{i=1}^{k} \sigma_i^2 U_i U_i^T = \sum_{i=1}^{k} \theta_i V_i, \quad V_i = U_i U_i^T.$$

We use a quadratic estimator $q = y^T A y$ with requirements:

(a) **Translation invariance:** $AX = 0$.

(b) **Unbiasedness:** $\operatorname{tr}(AV_i) = f_i$ for all i.

(c) **Minimum norm:** minimise $\|U^T AU - \Delta\|^2$ (Frobenius), where $\Delta = \mathrm{diag}(f_i/n_i \cdot I_{n_i})$.

Since $\mathrm{tr}(AU\Delta U^T) = \sum(f_i^2/n_i)$ is constant in A, minimising the norm is equivalent to minimising

$$\mathrm{tr}(UV_0AV_0A) = \mathrm{tr}(AV_0AV_0), \quad V_0 = UU^T = \sum V_i.$$

8.3 Solving the MINQUE Problem

Theorem 8.1 (MINQUE Solution)**.** *Let $R = V_0^{-1} - V_0^{-1}X(X^TV_0^{-1}X)^-X^TV_0^{-1}$. The MINQUE of $p^T\theta$ is achieved by $A_* = \sum_i \lambda_i RV_iR$, where $\lambda = S^{-1}f$ and S is the* MINQUE matrix *with entries*

$$S_{ij} = \mathrm{tr}(RV_iRV_j).$$

The estimator is

$$\hat{\theta} = S^{-1}q, \quad q_i = \hat{\varepsilon}^TV_0^{-1}V_iV_0^{-1}\hat{\varepsilon},$$

where $\hat{\varepsilon} = y - X\hat{\beta}$ with $\hat{\beta} = (X^TV_0^{-1}X)^-X^TV_0^{-1}y$.

Proof. We verify A_* is optimal.

Feasibility: A_* is symmetric and $A_*X = \sum \lambda_i RV_iRX = 0$ since $RX = 0$.

Optimality: For any feasible A, compute $\mathrm{tr}(AV_0A_*V_0)$. Using $AV_0R = A$ (since $AX = 0$):

$$\mathrm{tr}(AV_0A_*V_0) = \sum_i \lambda_i \,\mathrm{tr}(AV_0RV_iRV_0) = \sum_i \lambda_i \,\mathrm{tr}(AV_i) = \sum_i \lambda_i f_i,$$

which is constant in A. For the inner product $\langle A, B\rangle = \mathrm{tr}(AV_0BV_0)$:

$$\|A\|^2 = \|A - A_*\|^2 + \|A_*\|^2 - 2(\langle A, A_*\rangle - \langle A_*, A_*\rangle) + \ldots = \|A - A_*\|^2 + \|A_*\|^2 \geq \|A_*\|^2,$$

so A_* minimises the norm. The expression $q_i = y^TRV_iRy$ follows from $Ry = V_0^{-1}\hat{\varepsilon}$. ■

8.4 MINQUE Procedure: Step-by-Step

Step 1. Compute $\hat{\beta} = (X^TV_0^{-1}X)^-X^TV_0^{-1}y$ using the prior weights V_0.

Step 2. Compute residuals $\hat{\varepsilon} = y - X\hat{\beta}$.

Step 3. Compute $q_i = \hat{\varepsilon}^T V_0^{-1} V_i V_0^{-1} \hat{\varepsilon}$ for $i = 1, \ldots, k$.

Step 4. Compute $S_{ij} = \mathrm{tr}(RV_iRV_j)$ where $R = V_0^{-1} - V_0^{-1}X(X^TV_0^{-1}X)^-X^TV_0^{-1}$.

Step 5. Solve $S\hat{\theta} = q$ to obtain the MINQUE $\hat{\theta}$.

8.5 Examples

Example 8.1 (Ordinary Linear Model). In $(y, X\beta, \sigma^2 I)$ we have $k = 1$, $V_1 = I$, $\theta_1 = \sigma^2$, and $V_0 = I$. Then $R = I - P_X$,

$$S_{11} = \mathrm{tr}(R \cdot I \cdot R \cdot I) = \mathrm{tr}(I - P_X) = n - \mathrm{rk}(X),$$

$$q_1 = y^T(I - P_X)y = \mathrm{RSS}.$$

The MINQUE of σ^2 is

$$\hat{\sigma}^2 = \frac{q_1}{S_{11}} = \frac{\mathrm{RSS}}{n - \mathrm{rk}(X)},$$

the standard unbiased estimator.

Example 8.2 (One-Way Random Effects Model). Consider $y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$ for $i = 1, \ldots, a$, $j = 1, \ldots, n$, with $\alpha_i \sim (0, \sigma_A^2)$ and $\varepsilon_{ij} \sim (0, \sigma_e^2)$ all independent. In matrix form:

$$V = \sigma_A^2(I_a \otimes J_n) + \sigma_e^2 I_{an}, \quad V_1 = I_a \otimes J_n, \quad V_2 = I_{an}.$$

Take prior weights $w_1 = 0$, $w_2 = 1$, so $V_0 = I_{an}$ and $X = \mathbf{1}_{an}$, giving $R = I - \frac{1}{an}J_{an}$.

Computing the MINQUE matrix:

$$(I - \tfrac{1}{an}J_{an})(I_a \otimes J_n) = (I_a \otimes J_n) - \tfrac{1}{a}J_{an}.$$

$$\begin{aligned}
S_{11} &= \mathrm{tr}\Big[(I_a \otimes J_n - \tfrac{1}{a}J_{an})^2\Big] = n^2(a-1), \\
S_{12} &= \mathrm{tr}\Big[R(I_a \otimes J_n)R\Big] = n(a-1), \\
S_{22} &= \mathrm{tr}(R^2) = \mathrm{tr}(R) = an - 1.
\end{aligned}$$

Hence the MINQUE matrix is

$$S = \begin{pmatrix} n^2(a-1) & n(a-1) \\ n(a-1) & an-1 \end{pmatrix}.$$

The MINQUE of $(\sigma_A^2, \sigma_e^2)^T$ is $\hat{\theta} = S^{-1}q$.

Remark 8.2. Under normality, MINQUE coincides with MIVQUE (Minimum Variance Quadratic Unbiased Estimator), which minimises $\mathrm{Var}(y^T Ay) = 2\,\mathrm{tr}(AVAV)$ (using $AX = 0$) locally at the prior guess V_0.

Chapter 9

Summary of Key Results

The table below collects the main results for the General Linear Model $(y, X\beta, \sigma^2 V)$ with V known.

No.	Result
1	$k^T y$ is an LUE of $p^T\beta \iff \exists\, \ell$ s.t. $k^T y = \ell^T y$ a.s. and $X^T \ell = p$.
2	$k^T y$ is an LZF $\iff \exists\, \ell$ s.t. $k^T y = \ell^T y$ a.s. and $X^T \ell = 0$.
3	$(I - P_V)y = d$ (non-stochastic); admissible β must satisfy $(I - P_V)X\beta = d$.
4	Any LZF equals a linear combination of $(I - P_X)y$.
5	Any BLUE is uncorrelated with every LZF.
6	If $A\beta$ is estimable, the unique BLUE of $A\beta$ is $A\hat{\beta} = AX^-\hat{y}$, where $\hat{y} = [I - V(I - P_X)[(I - P_X)V(I - P_X)]^-(I - P_X)]y$.
7	$\mathcal{D}(\hat{y}) = \sigma^2[I - V(I - P_X)[(I - P_X)V(I - P_X)]^-(I - P_X)]V$.
8	$e = y - \hat{y};\ \mathcal{D}(e) = \sigma^2 V(I - P_X)[(I - P_X)V(I - P_X)]^-(I - P_X)V$.
9	The LZF space has dimension $d = \mathrm{rk}(X : V) - \mathrm{rk}(X)$.
10	Under normality, $R_0^2/\sigma^2 \sim \chi_d^2$, giving $\hat{\sigma}^2 = R_0^2/d$.

Notation Reference

Symbol	Meaning
$\mathcal{C}(A)$	Column space of A
P_X	Orthogonal projector onto $\mathcal{C}(X)$
$\mathrm{rk}(A)$	Rank of A
A^-	A generalised inverse of A
A^T	Transpose of A
$\mathcal{D}(z)$	Dispersion (covariance matrix) of random vector z
LUE	Linear Unbiased Estimator
LZF	Linear Zero Function
BLUE	Best Linear Unbiased Estimator
MINQUE	Minimum Norm Quadratic Unbiased Estimator
SUR	Seemingly Unrelated Regressions